Personal-Data Discovery & Privacy Evidence Inventory

AIORI Hackathon 3 • Category: Cybersecurity

Problem Description

Modern data protection frameworks require organizations to identify stored Personally Identifiable Information (PII) and firmly link it to declared business purposes and retention schedules. Over time, personal data fields inevitably accumulate across unindexed database tables and loose spreadsheets without clear or maintained documentation.

This creates a significant compliance gap. Organizations face the critical challenge of discovering this siloed data, categorizing it, and evaluating it against legal constraints without manual, error-prone audits.

The core problem addressed by this project is building an automated scanner capable of inspecting these datasets, mapping them to existing metadata, and surfacing technical evidence for human review.

Background

Regulatory environments worldwide have shifted from passive data collection to active data governance. Organizations act as data fiduciaries and are responsible for knowing exactly what data they hold, why they hold it, and when it should be deleted. Manually tracking data across disparate CSVs and structured SQL databases is highly prone to human error, leading to compliance violations and unauthorized data retention.

Our solution bridges the gap between raw data storage and compliance reporting. By providing a technical evidence inventory, we empower data protection officers to make informed legal interpretations based on accurate, automatically generated technical maps of the organization's data footprint.

Literature & RFC Review

A detailed study of existing proposals and standards reveals a consistent emphasis on the principles of purpose limitation and storage limitation.

Reference Standard: MeitY Data Protection Framework

This framework outlines the standard regulatory expectations for data fiduciaries in India. It mandates strict adherence to purpose declaration and data retention limits.

Existing solutions often attempt to make automated legal decisions, which is explicitly discouraged. Our technical evidence generator strictly focuses on discovery and metadata mapping, leaving lawful-basis determinations to human operators.

Proposed Solution & Optimization

We propose a highly configurable Python-based scanning utility that ingests synthetic CSV files and structured SQL databases. Using a configurable rules engine, it identifies candidate personal-data fields across 5 predefined data classes. It cross-references these fields with a metadata registry to flag missing purpose or retention data.

Operational Metric	Before (Manual Audit)	After (Automated Tool)
Discovery Process	Manual inspection of CSVs and SQL schemas.	Automated scanning across 5 defined PII classes.
Metadata Linking	Disconnected spreadsheets prone to desynchronization.	Programmatic linking to purpose and retention metadata.
Gap Analysis	High risk of undetected missing compliance information.	Automatic flagging of missing data with precision reporting.

This optimization reduces audit times significantly while improving the precision and recall of personal data discovery across unindexed environments.

Architecture Design

The system architecture separates data ingestion, classification, and reporting into distinct modules, ensuring maintainability and clear bounds between technical scanning and compliance reporting.

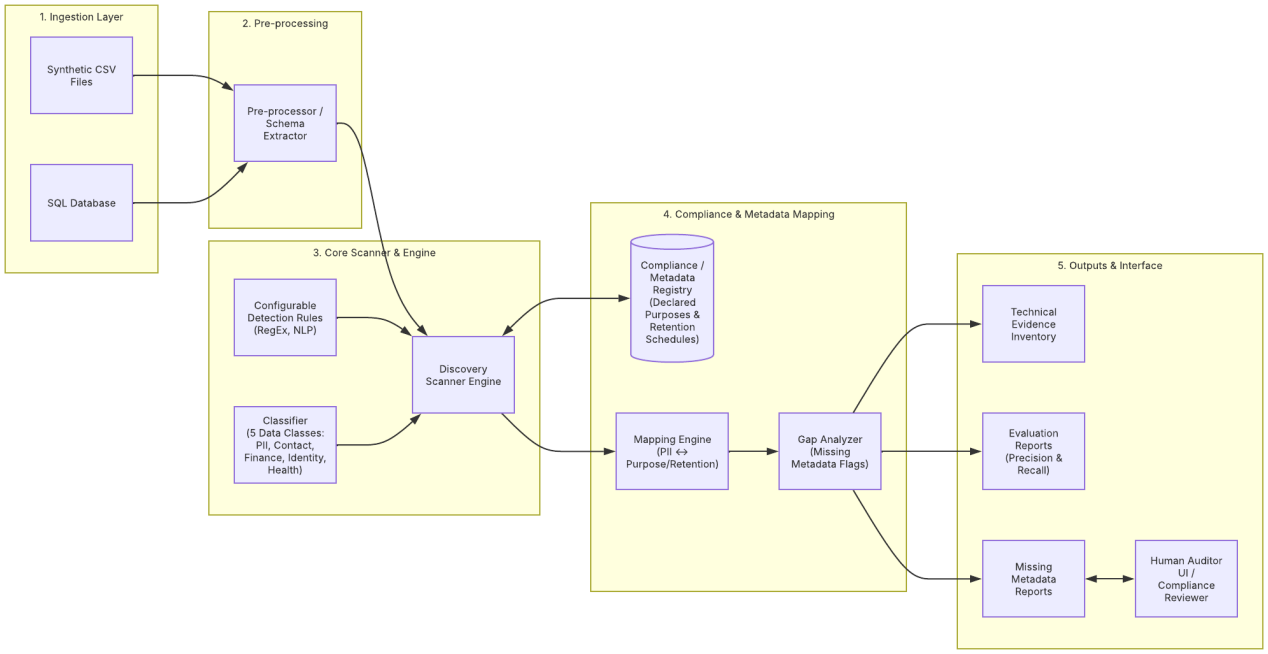


Figure 1: High-level System Architecture illustrating component separation.

Data Flow Diagram

Data is extracted from the source, passed through the classification engine, joined with metadata, and compiled into an actionable inventory report.

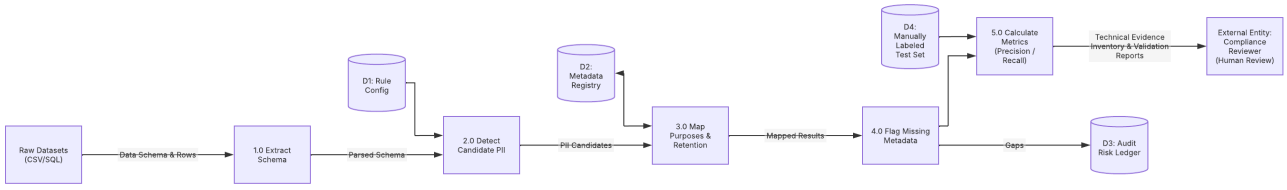


Figure 2: Data Flow illustrating ingestion, rules application, and metadata joining.

The flow guarantees that no data is permanently modified or moved; the scanner strictly generates read-only metadata reports and evaluates them against the manually labeled test set for precision and recall metrics.

Deployment Timeline

The implementation follows an iterative deployment strategy to ensure continuous mentor oversight via the GitHub repository.

Phase / Milestone	Key Deliverables & Workflow Actions
Phase 1: Environment & Integration	Repository setup, collaborator transfer (aiori-hackathon), and establishment of the CSV/SQL data ingestion pipeline.
Phase 2: Rules Engine Development	Development of configurable detection rules for the 5 candidate personal-data classes. Initial evaluation against the small manually labeled test set.
Phase 3: Metadata Mapping & Analysis	Integration of purpose and retention metadata linking. Implementation of logic to identify and flag missing compliance information.
Phase 4: Reporting & Demonstration	Generation of the final privacy evidence inventory report, precision/recall evaluation output, and preparation of the prototype demonstration with the required pseudocode snippet.

Bibliography

• **[1] Ministry of Electronics and Information Technology (MeitY).** "Data Protection Framework." Available: https://www.meity.gov.in/data-protection-framework.

• **[2] AIORI-3 Hackathon Challenge Brief.** "PS-019: Personal-Data Discovery and Privacy Evidence Inventory."